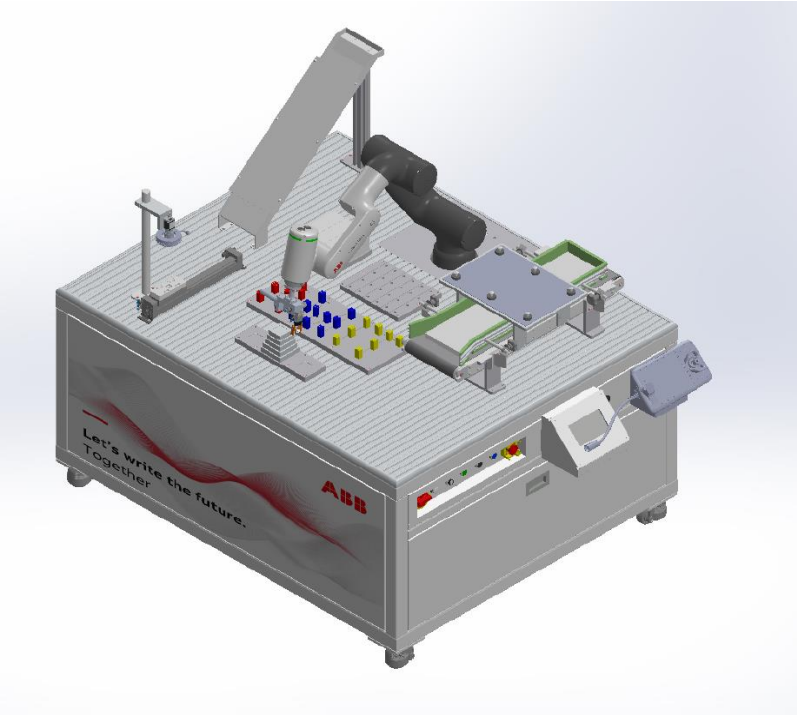
ABB ROBOTICS VIETNAM

RMIT_EDUCATION CELLS

Operating Manual



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1. PURPOSE AND BASIC DESCRIPTION

The Gofa Cobot Education Cell is a robotic training station designed to provide a safe and engaging hands-on learning experience. This collaborative robot (cobot) system allows users to program and operate the robotic arm through a user-friendly interface. The station features a range of safety mechanisms, including sensors and emergency stop functions, to ensure a secure learning environment.

The Gofa Cobot Education Cell is a robotic training station designed to provide an immersive learning experience in automation and programming. This collaborative robot (cobot) system allows users to design and execute a variety of hands-on tasks, including object pickup and placement, pyramid building, and block arrangement to spell out 'RMIT'. The intuitive interface enables users to easily program the robotic arm to carry out these demonstrations. Comprehensive safety features, such as sensor-based collision detection and emergency stop controls, ensure a secure environment for exploration and experimentation. Whether you are new to robotics or looking to expand your automation skill set, the Gofa Cobot Education Cell offers an engaging platform to develop practical competencies in a collaborative robot system. Follow the step-by-step instructions in this manual to get started operating, programming, and maintaining the Gofa Cobot Education Cell.

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1.1. SYSTEM DETAILS

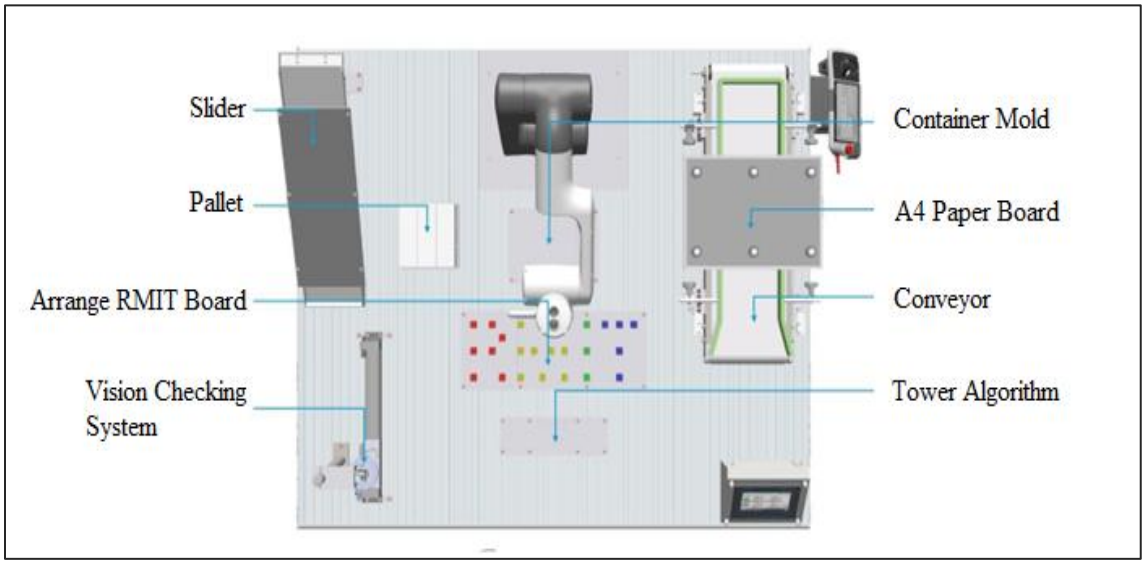


Figure 1. Layout of the top of the station

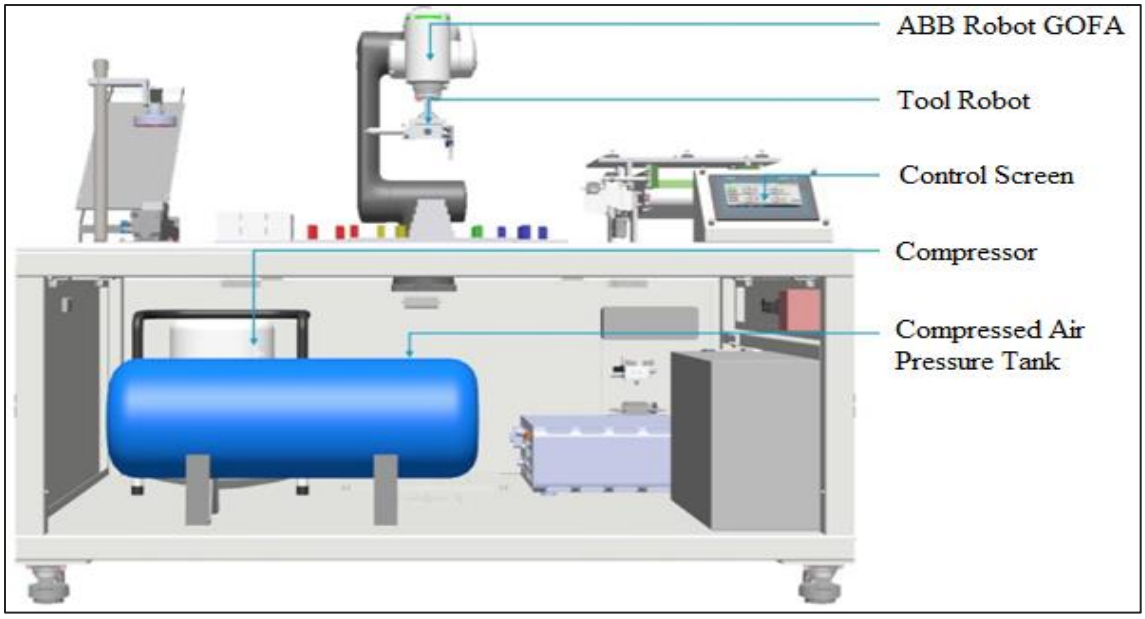


Figure 2. Inside and above layout of the station

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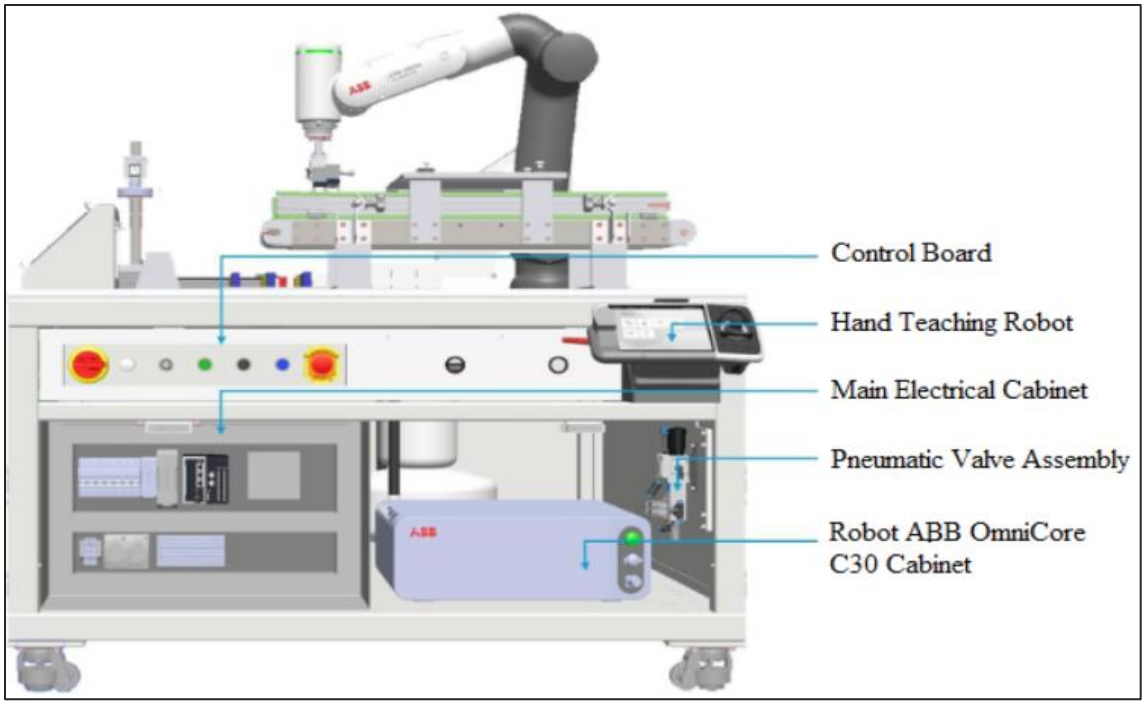
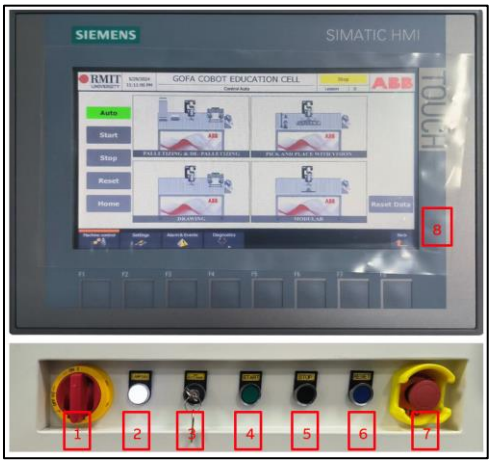


Figure 3. Layout on the left side of the station

1.2. USER INTERFACE

This section will describe the basic functionality and operation of each component on the main control panel.



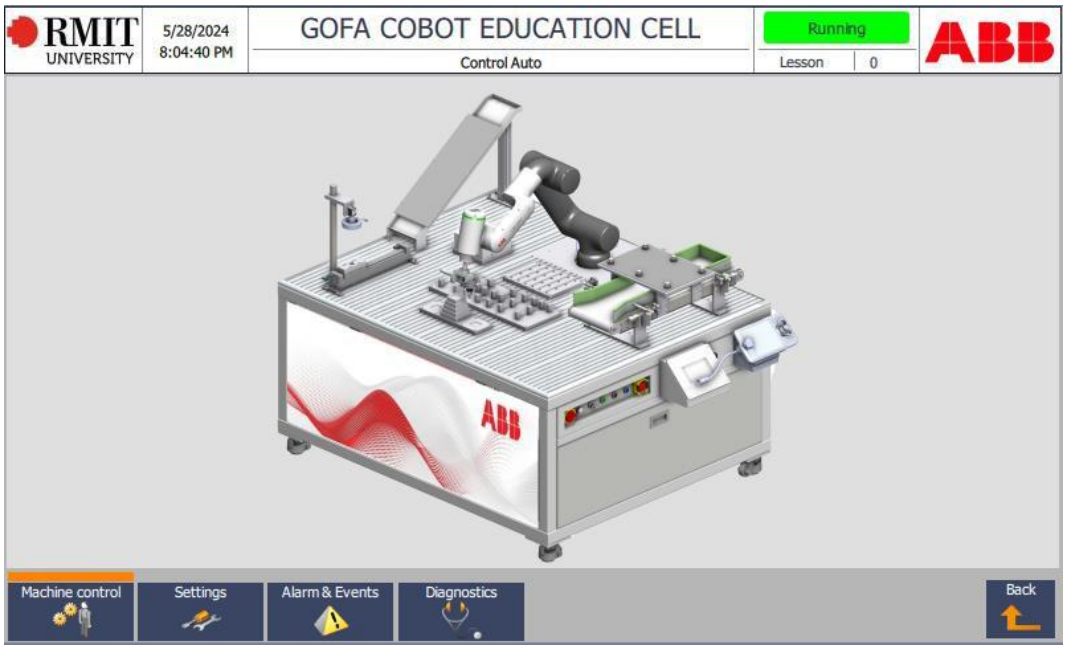
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No	Label	Describe
1	MAIN SWITCH	Open the electrical cabinet’s source.
2	DC POWER	Displays the DC power status of the station.
3	MANUAL/OFF/AUTO	A key will be used to switch the station's usage mode. In MANUAL mode, the station equipment can be controlled. In AUTO mode, the Robot runs the selected programs. In OFF mode, the stop station operates, cannot be adjusted control all equipment.
4	START	Press the START button in automatic mode to let the Robot run the selected program.
5	STOP	Press the STOP button to stop the station while it is running.
6	RESET	RESET button to reset the station's status when an error occurs.
7	EMO	Emergency Stop Button (Emergency Machine Off).
8	Touchscreen	The HMI display displays the control buttons and related information.

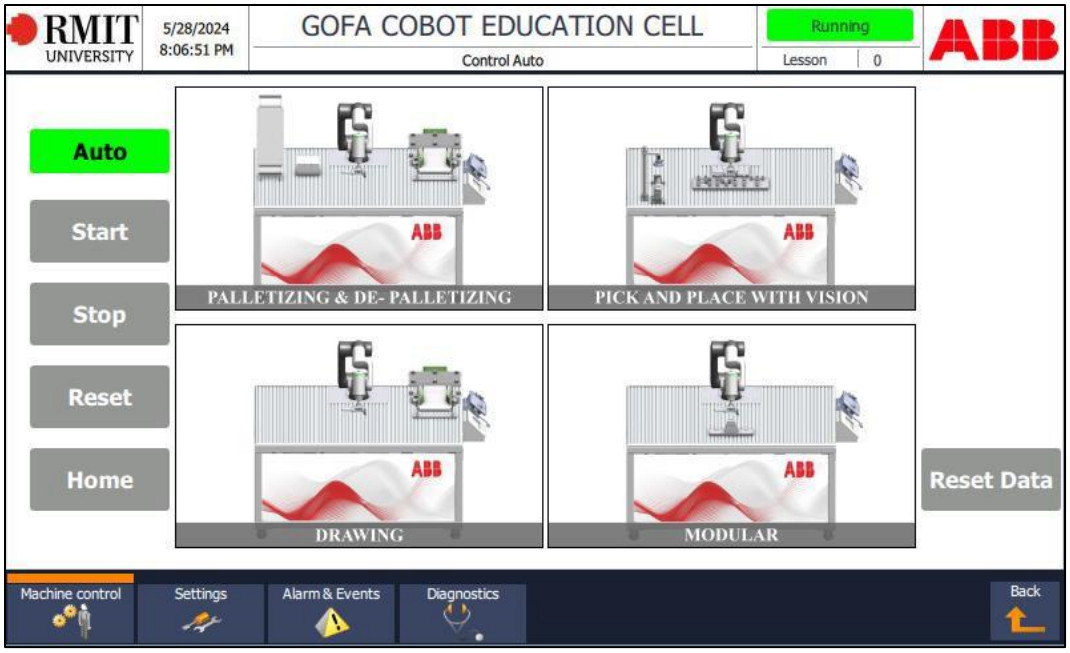
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1.3. HMI MAIN SCREEN

In this image, you can press the button below to go to the viewing screen you want to go to.



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1.4. Machine Control Screen

From this screen, users can visually see the dynamic status of the system, such as: emergency stop status, manual/automatic status of the system, and select a program for the Robot.

No	Label	Describe
1	AUTO/MANUAL/OFF	Displays the operating status of the system. **AUTO:** The system is in automatic mode. **MANUAL:** The system is in manual mode. **OFF:** The system is not working.
2	Start	Operating System
3	Stop	Pause System
4	Reset	Remove system errors
5	Home	About the original state of the robot
6	Reset Data	Clear data

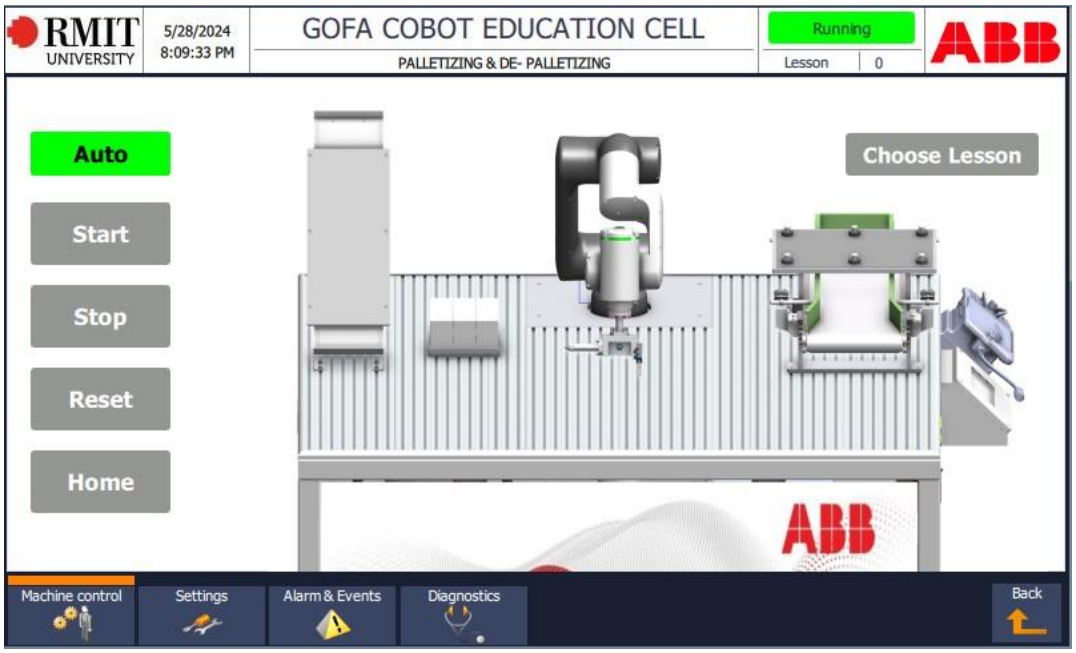
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1.4.1. Palletizing & De-Palletizing

Palletizing is a program for the Picking Robot from the slider placed on the top of the conveyor, when the product runs to the end of the Picking Robot conveyor and is placed in the position of the Pallet in an even or odd position.

De-Palletizing is a robot program that picks up products from pallets in an even or odd position and places them on the top point of the slider.

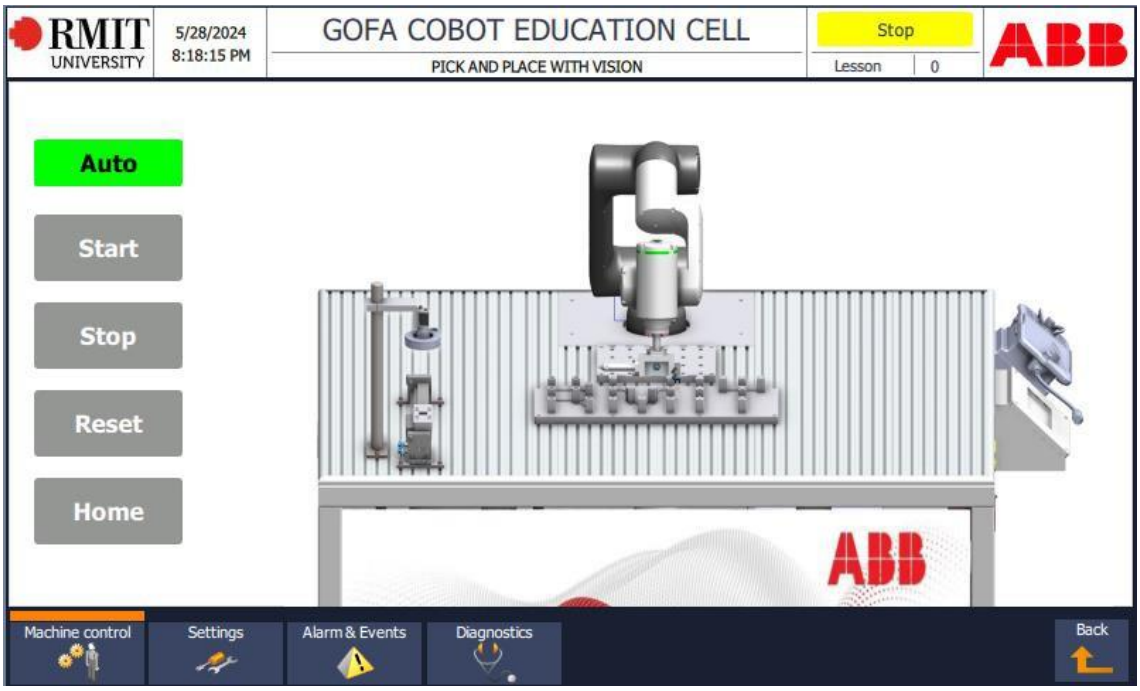
Choose Lesson to select a process.



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1.4.2. Pick and Place with Vision

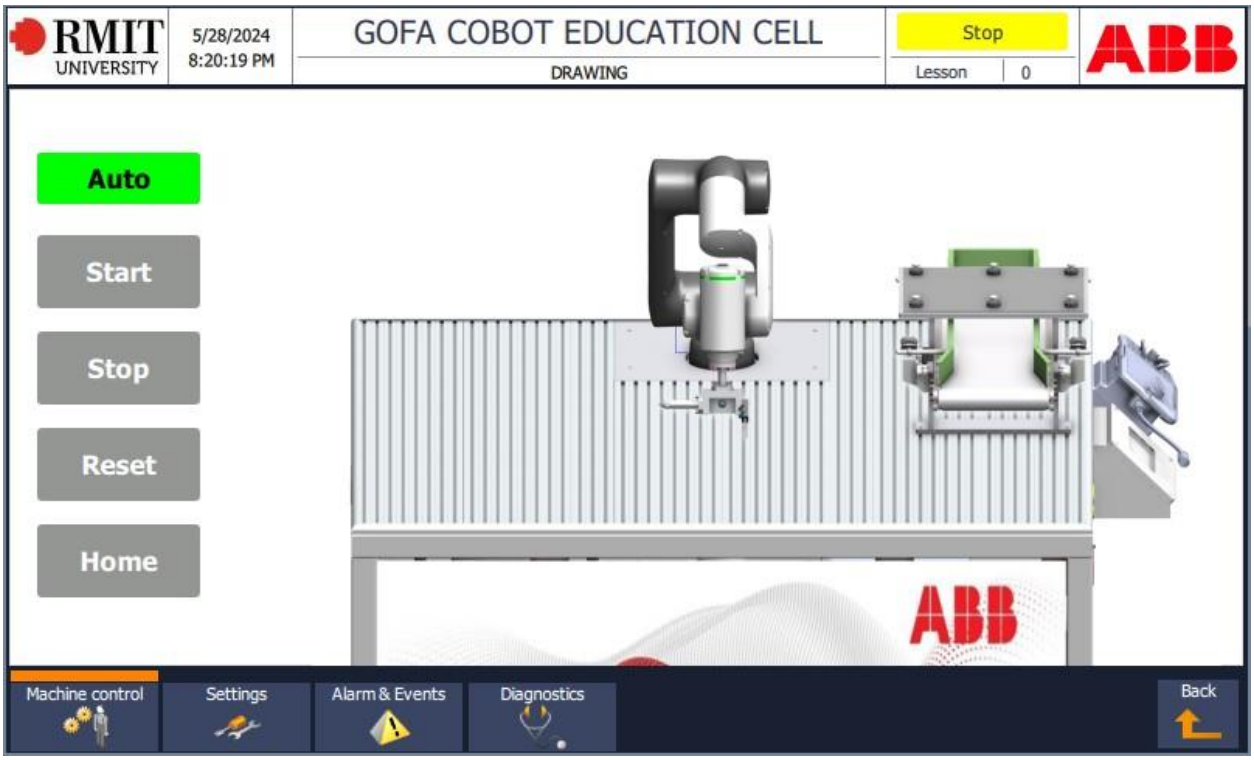
The robot picks up the rod from the mold and puts it on the vision test system, then receives the shape and color of the rod again, picks it up and puts it into the hole on the "RMIT" board, completing the process, the robot picks up the bars and puts them back into the mold.



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1.4.3. Drawing

The robot draws the letters "R M I T" on A4 paper.

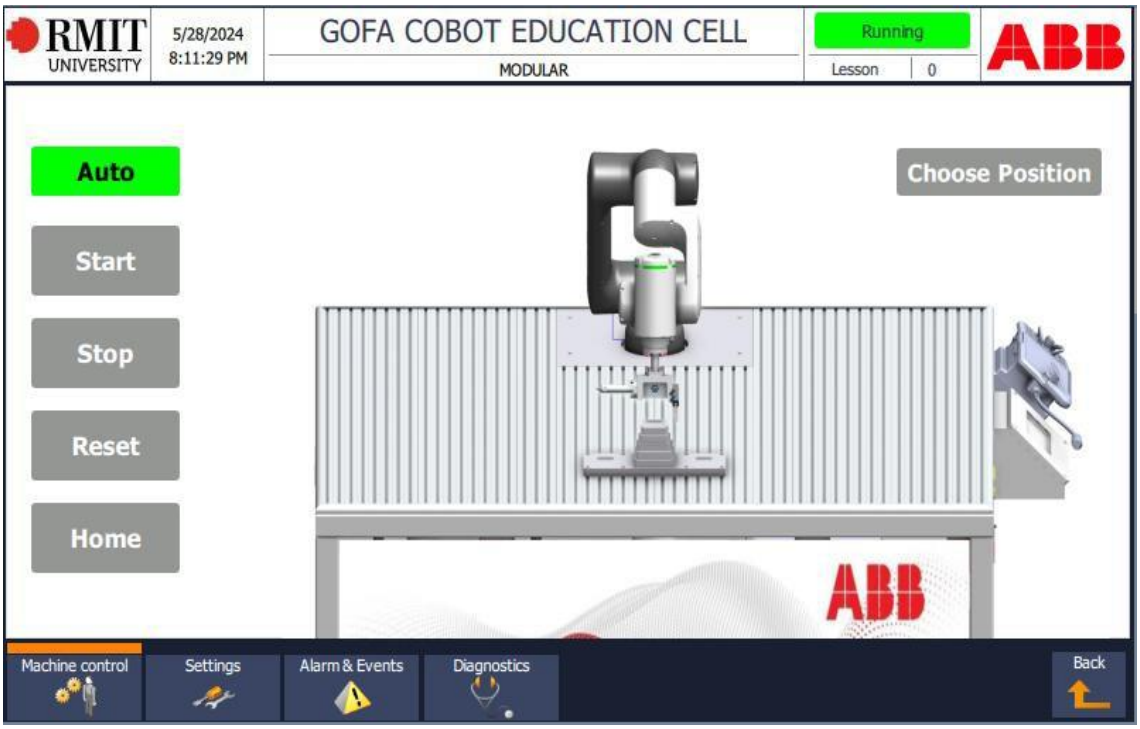


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1.4.4. Modular

The robot picks up and places according to the Hanoi Tower algorithm, depending on the location.

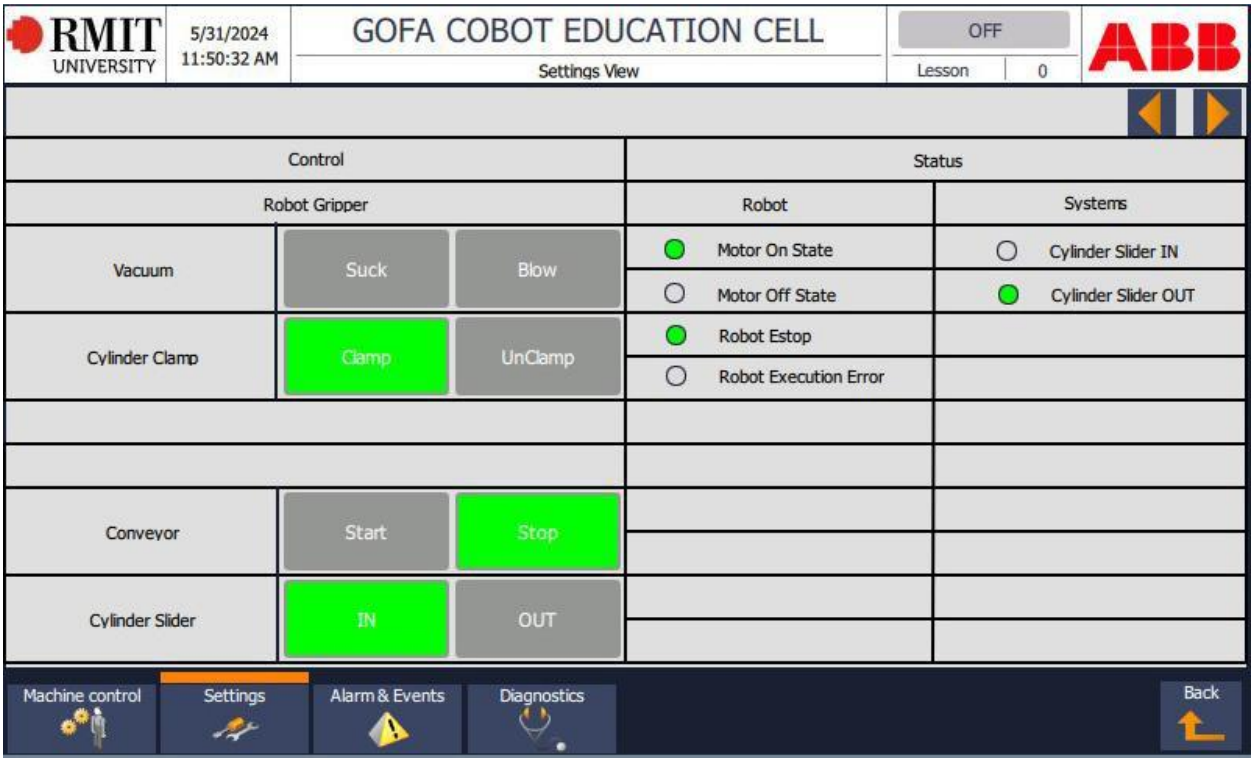
Choose Position to confirm the original location of the Tower.



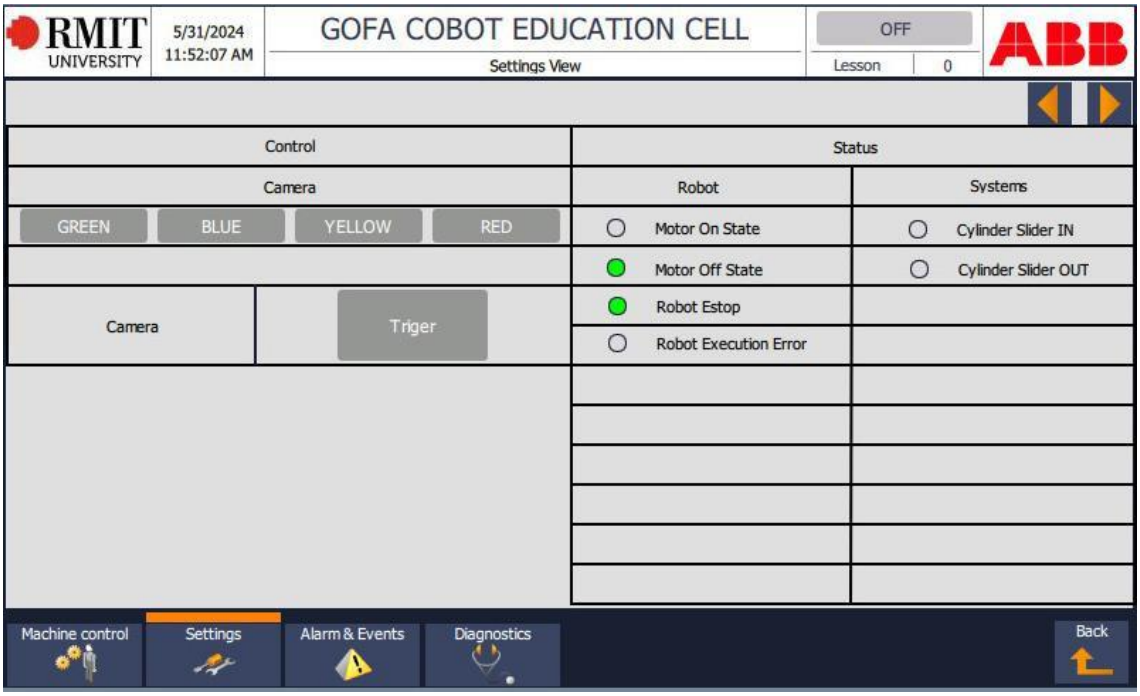
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1.5. SETTING SCREEN

On the Settings page, users can open and close cylinders, vacuums, conveyors, cameras, robots, sensors, cameras, etc.



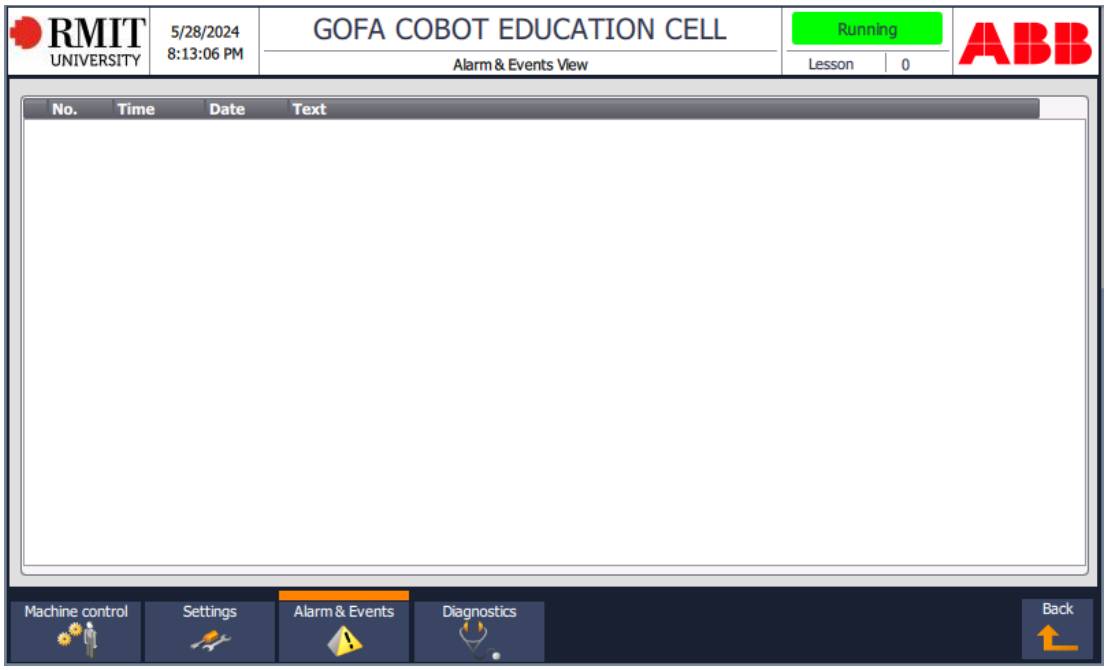
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1.6. ALARM & EVENT DISPLAY

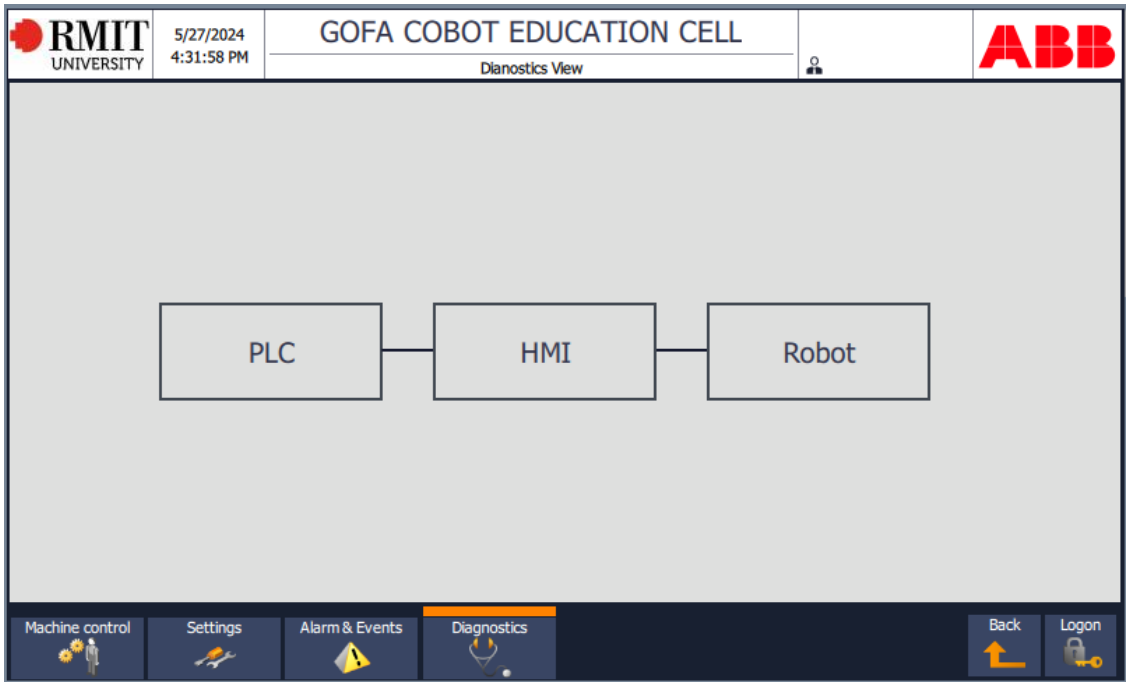
On the Alarm page, an error warning. Displays the current faults of the system. Display the time and content of the fault for the operator to handle the fault.



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1.7. SCREEN DIAGNOISTICS

In the Diagnostics view, the interface displays the transmission status of the Profinet communication network.

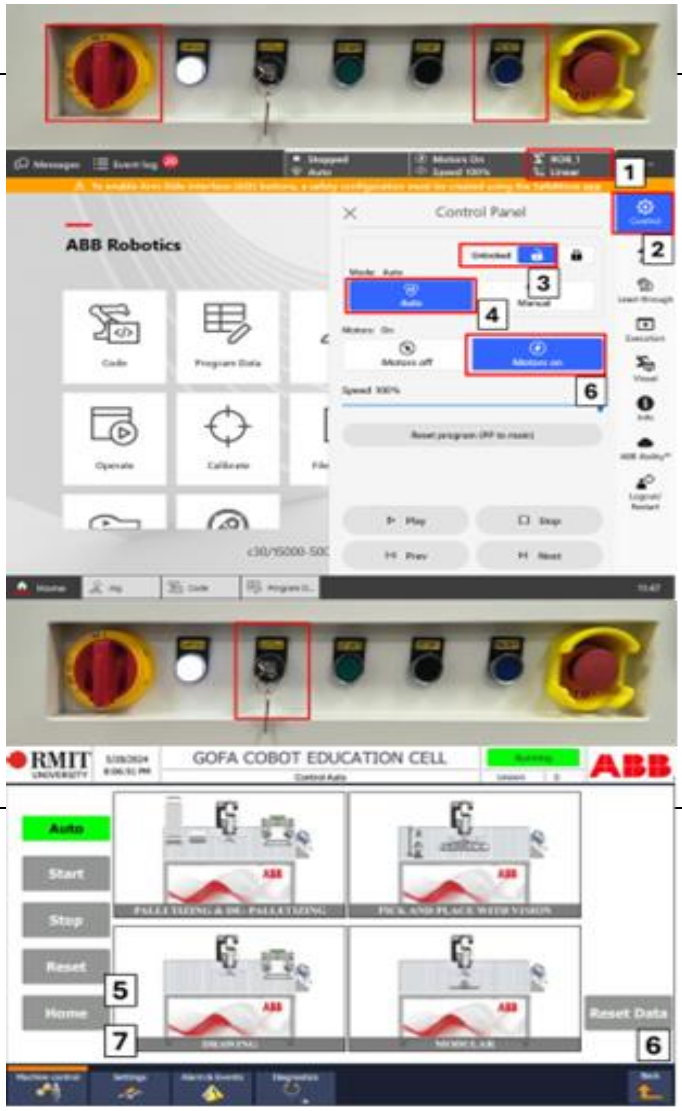


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2. OPERATING THE STATION

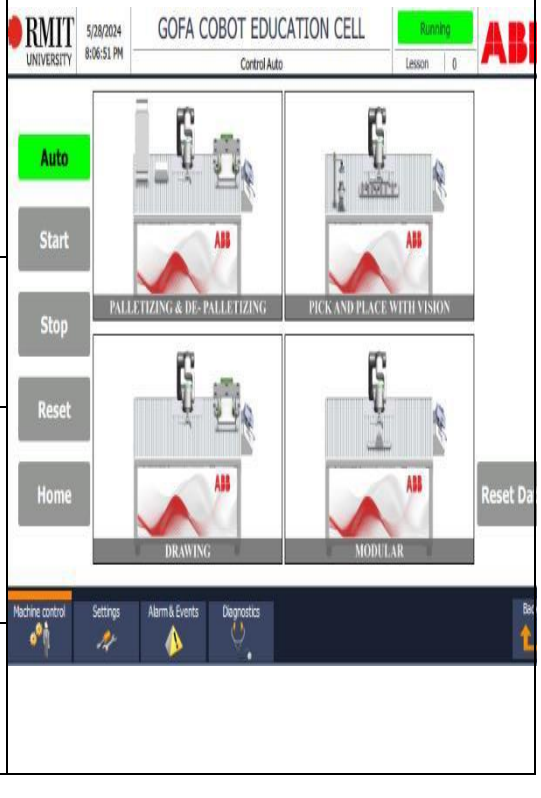
Note: Follow one step at a time to avoid possible errors.

2.1. STATION OPENING PROCEDURE

Step	Action	Image
1	Open the electrical cabinet power switch	
2	Press the RESET button on the control cabinet	
3	Switch the Robot to automatic mode: • Select the Robot control screen • Select Control • Unlocked • Switch Robot to Auto • Select Acknowledge to confirm • Open Motor ON	
4	Switch the system to automatic mode by switching the lock to AUTO mode.	
5	Reset	
6	Reset Data	
7	Home – Wait In Home	

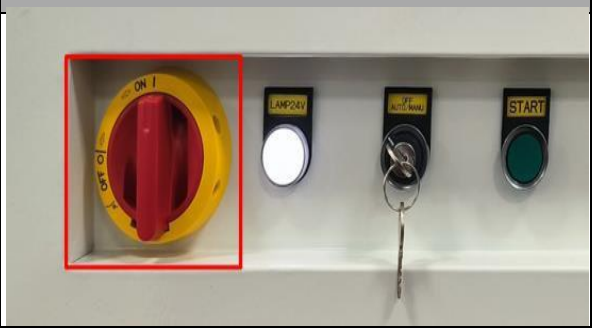
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Select the program on the HMI and run Auto

Step	Action	Image
1	Choose a task	
2	Select Sub-Module Lesson 1: *Palletizing or Depalletizing* Lesson 4: Choosing the initial location: *Left, Center or Right*	
3	The Ready status appears **Press Start**	
4	When the run is complete, Complete appears **Reset** **Reset Data**	
5	Home – Wait In Home Go back to step 1.	

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2.2. STATION SHUTDOWN PROCEDURE

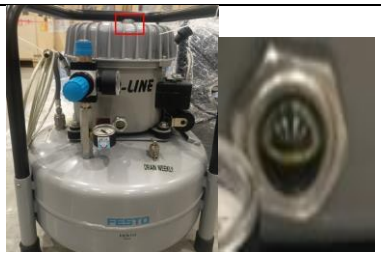
Step	Action	Image
1	Bringing Robot to Home Home - wait for In Home	
2	Turn off the electrical cabinet power switch	

2.3. AIR COMPRESSOR OPERATION PROCEDURE

Step	Action	Image
1	Plug in the power for air compressors	 
2	ON Air Compressor Power (On 4 Bar - 8 Bar) (OFF 8 Bar)	
3	Open Valve Pneumatic	

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2.4. AIR COMPRESSOR MAINTENANCE PROCEDURE

Step	Action	Image
1	- Discharge waste for the machine (1 week / 1 time)	
2	- Oil Filling for Air Compressor	

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3. ADDITIONAL INFORMATION

3.1. SYSTEM SAFETY

3.1.1. Safety Overview

The user must read the safety instructions before working with the robot to prevent accidents when using it. The robot's safety devices and programs must be checked before use.

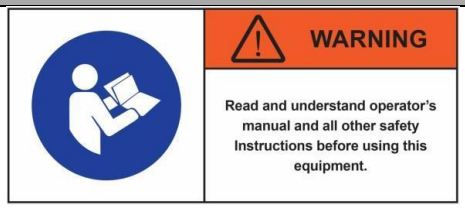
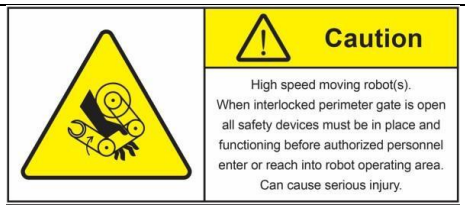
Dangerous	The warning indicates an imminent danger, which, if not avoided, is very likely to result in serious injury heavy.
Alert	Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury
Cautious	Indicates a potentially hazardous situation, which, if not avoided, could result in minor or moderate injury.
Notes Heavy	Featured items that should not be ignored to ensure the station is active smooth movement.

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3.1.2. Safety Notice

Important warning labels are defined by the following color codes:

General warning labels.

Hazard Labels	Action
	Level: Dangerous Action: read the instructions carefully before use Location: HMI Cabinet
	Level: Dangerous Action: The device still has power after EMO is activated Location: Electrical Cabinet
	Level: Dangerous Action: Turn off the power and lock before repairing Location: Electrical cabinet
	Level: Caution Action: Do not touch the conveyor belt while in operation. Location: Conveyor Belt
	Level: Caution Location: Robot

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3.1.3. Safety Requirements

Individuals working in the Robot station must comply with the following regulations:

1. The work area must be clean.
2. The protective devices used, the safety signals, and other safety devices must not be damaged. Never work on a safety device or an appliance with a defective safety ring.
3. All individuals concerned must be fully aware of the location of all station emergency stop switches.

3.1.4. Emergency stop button

The emergency stop button is integrated on the robot teaching arm, and on the control panel. This emergency stop button must only be pressed when an urgent critical problem occurs during use. In case the system is stopped at a non-critical problem, please use the **STOP** button instead. The **STOP button** will pause the station's operation without affecting the robot's operating procedure.

Emergencies when the robot is in operation have the potential to collide, damage the robot tool or collide with humans. In these cases, we need to press the EmergencyMachine Off button immediately to avoid unfortunate things from happening.

Emergency stop button on the teaching hand	**Control Panel**
	

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3.1.5. Recover after pressing the emergency stop button

Perform the operations below to reuse the station after pressing the Emergency MachineOff button

Step 1. Identify the problem and solve the problem that occurs.

Step 2. The emergency button release is activated by turning the knob counterclockwise.

Step 3. Press the **RESET button** on the control cabinet.

Step 4. Press the **Motor On button** to reset the status of the Robot.

Step 5. From the HMI screen, go to the **Machine Control page** and press the **Home button**.

Step 6. Confirm that the problem has been solved and check the usage investigation.

Step 7. Press the **START button** on the control cabinet to allow the station to continue operating.

3.2. ERROR HANDLING

3.2.1. Error: Cylinder does not Receive Sensor Signal

Check the fault cylinder name on the HMI ERROR Screen.

Use the screwdriver to adjust the sensor on the cylinder body, moving up/down until the sensor lights up.

3.2.2. Error: Robot Fails to Change to Auto Mode

Switch the Robot to Auto mode

3.2.3. Error: Robot can not move to Home Position

Move the Robot to a safe position and run again